CIRCULAR RESOURCE RECOVERY // INDUSTRIAL WORKSPACE RUNBOOK

SYSTEM USER MANUAL

VERSION 1.0 // AY 2025–2026

Operational Guide for B2B By-Product Exchange, Real-Time Reverse Auction & SPCB Form 10 Compliance

Comprehensive step-by-step documentation for industrial procurement teams, plant dispatch engineers, environmental auditors, and corporate administrators operating on the TRASURE platform.

AUTHOR SPECIFICATION & TECHNICAL DOSSIER

CONFIDENTIAL ACADEMIC RELEASE

Niv J. Sorathiya

Roll Number: 24BCB037

Branch: Computer Science and Business Systems (CSBS)

Academic Cohort: 5th Semester | Division 1 | Group J2

Course: Software Engineering (20CS301T / 20CS301P)

College Email: 24bcb037@sot.pdpu.ac.in

Personal Email: niv.sorathiya@gmail.com

Live Platform: https://trasure-phi.vercel.app **Source Code: https://github.com/nivsorathiya-web/TRASURE**

12,500+	₹8.2 Cr+	150+	100%
TONNES DIVERTED	COMMERCIAL LIQUIDITY	VERIFIED PLANTS	SPCB AUDIT TRAIL

01 TABLE OF CONTENTS & OPERATIONAL ARCHITECTURE

System Architecture & User Roles

TRASURE operates as a closed-loop B2B exchange connecting industrial by-product generators with secondary consumers across Indian manufacturing corridors.

User Manual Modules

MODULE	TITLE & KEY SCOPE	TARGET ACTOR
Mod 1	Corporate Access & Workspace Authentication	All Personnel
Mod 2	Circular Material Discovery & Landed Cost	Procurement
Mod 3	GIS Highway Transit Routing & Logistics Map	Logistics Lead
Mod 4	Real-Time Reverse Auction & Anti-Sniping	Traders & Buyers
Mod 5	SPCB Form 10 Compliance & Manifest Vault	Auditor & Safety
Mod 6	Verified Plant Directory & Administration	Plant Dispatch
Mod 7	Troubleshooting, FAQ & Support Escalation	Technical Ops

Role Access Matrix

Platform capabilities partition dynamically based on active corporate credentials:

Sourcing Buyer: Accesses catalog, compares delivered landed costs, initiates reverse auction tenders, and downloads tax invoices.

Listing Seller: Lists secondary by-products, sets ex-plant reserve floor, competes in dynamic counter-bidding, and issues delivery notes.

Regulatory Auditor: Inspects SHA-256 cryptographic manifests and verifies SPCB hazardous consent codes.

https://trasure-phi.vercel.app/console

LIVE VERIFIED



Figure 1: Project TRASURE 3-Step Symbiosis Operations — Browse & Filter, Live Reverse Auction, and Form 10 Compliance Generation.

Default Test Enterprise Profile

Corporate: **Ambuja Cements Ltd (Ahmedabad Unit)** | PIN: 1234 | Mode: **Sourcing Buyer**

PRE-CONFIGURED ACCESS

02 MODULE 1: ACCESS CONTROL & SECURITY PROTOCOL

Corporate Authentication & Workspace Login

Enterprise access to TRASURE utilizes a streamlined, credentialed portal designed to prevent unauthorized trading while providing instant role-based access for verified industrial plants.

Step-by-Step Login Sequence

1. **Launch Portal:** Navigate to https://trasure-phi.vercel.app in any modern browser.
2. **Select Role Mode:** Click the **Sourcing Buyer** or **Listing Seller** tab based on current operational context.
3. **Choose Plant Profile:** Select your registered entity from the dropdown (e.g., *Ambuja Cements Ltd*).
4. **Enter Access PIN:** Input your 4-digit security PIN (default credential: 1234).
5. **Outbid Notifications:** Ensure *"Enable WhatsApp bidding & outbid alert updates"* is checked for real-time tender feeds.
6. **Authorize:** Click **Access Corporate Console →** to enter the dashboard.

Security Rules & Session Management

- **Brute Force Lockout:** 5 consecutive invalid PIN attempts triggers a 15-minute workstation freeze.
- **Session Isolation:** Corporate context persists in secure browser memory (`localStorage`) with auto-expiry.
- **Audit Logging:** Every authentication event records client IP, UTC timestamp, and corporate authorization hash.
- **SPCB Sync:** Only plants holding an active SPCB consent-to-operate certificate are permitted to bid on hazardous material lots.

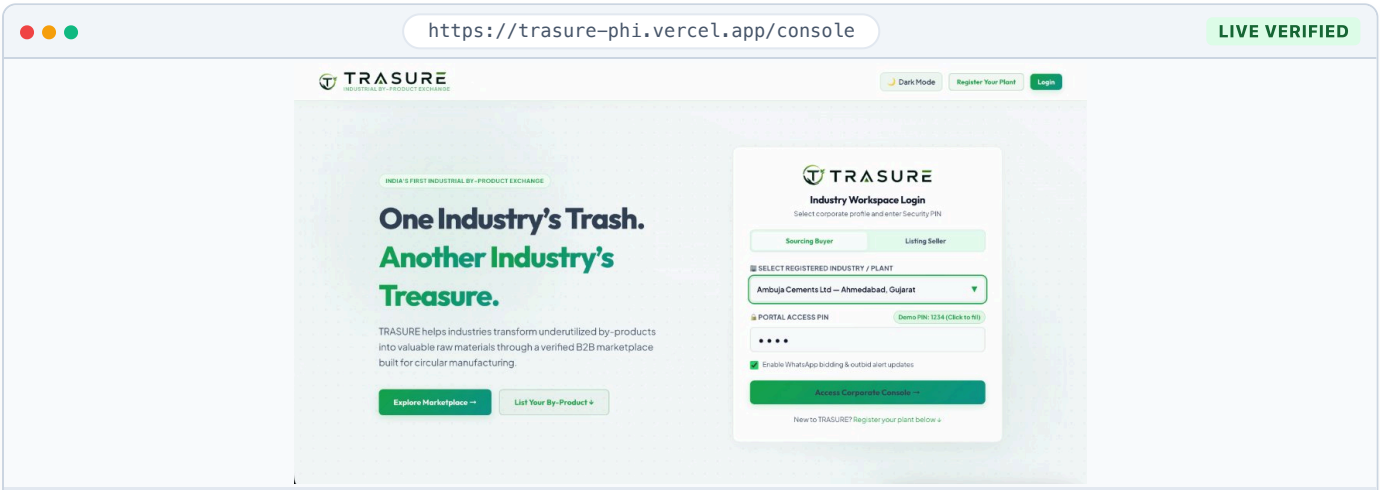


Figure 2: Corporate Workspace Login Modal — Sourcing Buyer / Listing Seller selection with Ambuja Cements profile and PIN 1234 verification.

Verification Note: Upon successful login, the system loads the plant's geographic coordinates, receiving hub address, and active material procurement requirements into client memory.

03 MODULE 2: MATERIAL DISCOVERY & COST OPTIMIZATION

Circular Sourcing Catalog & Landed Cost Engine

The TRASURE discovery catalog ranks available industrial by-products not by arbitrary list prices, but by true **Delivered Landed Cost** computed directly to the buyer's factory gate.

https://trasure-phi.vercel.app/console LIVE VERIFIED

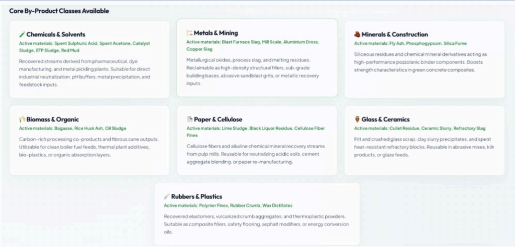


Figure 3: By-Product Classification Grid — Categorized industrial waste streams including Chemicals, Metals, Minerals, and Biomass.

https://trasure-phi.vercel.app/console LIVE VERIFIED

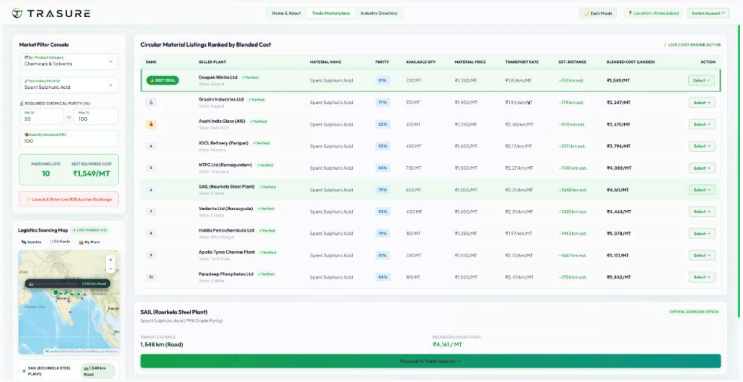


Figure 4: Circular Marketplace Console — Multi-parameter category filters, 10 active industrial listings, and dynamic delivered cost ranking.

Landed Cost Formula

```
Landed Cost = Ex-Plant Rate + (Road Dist × ₹2.01/km/MT) + 2% Clearing
```

Eliminates post-purchase freight surprises by accounting for highway toll corridors and axle weight limits.

Active By-Product Classes

Fly Ash: NTPC Vindhyachal (₹650/MT) | **Blast Furnace Slag:** SAIL Rourkela (₹1,200/MT) | **Spent Acid:** Deepak Nitrite (₹1,350/MT) | **Phosphogypsum:** Paradeep (₹420/MT).

04 MODULE 3: GEOGRAPHIC TRANSIT ROUTING

GIS Highway Transit Routing & Freight Optimization

Real-time spatial analysis calculates exact road mileage and estimated heavy-vehicle transit hours using high-precision OpenStreetMap road networks.

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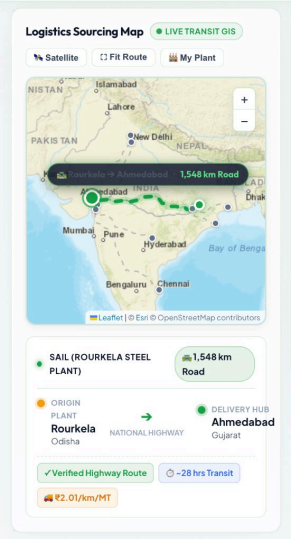


Figure 5: Live GIS Transit Routing — SAIL Rourkela to Ahmedabad (1,548 km Road Route, ~28 hrs transit, ₹2.01/km/MT).

Routing Engine Specification

- **Router API:** Open Source Routing Machine (OSRM) driving profile.
- **Highway Corridors:** Prioritizes National Highway (NH) 4-lane and 6-lane expressways over state roads.
- **Haversine Fallback:** Automatic mathematical great-circle fallback if offline: $d = 2R \times \arcsin(\sqrt{(\sin^2(\Delta\varphi/2) + \cos\varphi_1 \cos\varphi_2 \sin^2(\Delta\lambda/2))})$
- **Transit Calibration:** Assumes commercial heavy truck average velocity of 55 km/h with mandatory driver rests.

Corridor Freight Economics

CORRIDOR ROUTE	DISTANCE	TRANSIT	FREIGHT/MT
Rourkela → Ahmedabad	1,548 km	~28 hrs	₹3,111
Vindhyachal → Vadodara	1,120 km	~20 hrs	₹2,251
Hazira → Morbi	385 km	~7 hrs	₹774
Dahej → Surat	115 km	~2.5 hrs	₹231

Driver GPS Check-In: Transporters scan QR codes at highway toll booths to log checkpoint timestamps directly into the SPCB Form 10 manifest.

05 MODULE 4: DYNAMIC PRICE DISCOVERY

Real-Time Reverse Auction & Anti-Sniping Floor

TRASURE's real-time auction engine enables buyers to tender procurement contracts while pre-vetted industrial sellers submit downward (decremental) competitive counter-bids.

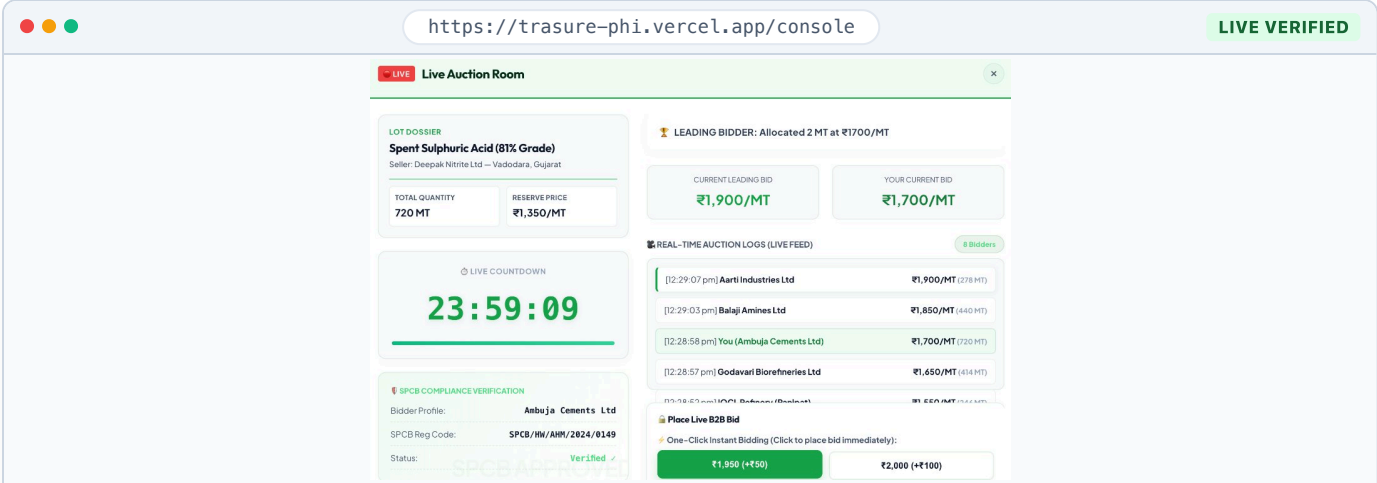


Figure 6: Live Auction Room — Spent Sulphuric Acid tender (720 MT, ₹1,350/MT reserve), 23:59:09 countdown, live bid logs, and SPCB verification.

Bidding Protocol & Rules

- **Reserve Ceiling:** Bids must strictly undercut the current leading rate.
- **Quick-Bid Increments:** One-click counter-bidding at `–₹50/MT` or `–₹100/MT`.
- **Broadcast Feed:** WebSocket/Event-stream updates active leader instantly across all connected terminals.

60-Second Anti-Sniping Rule

If a competitive counter-bid is submitted when the countdown clock displays **≤ 60 seconds**, the engine automatically adds **+60 seconds** to the tender timer, ensuring all participants fair response time.

06 MODULE 5 & 6: REGULATORY GOVERNANCE & DIRECTORY

SPCB Compliance Vault & Verified Plant Directory

Full legal alignment with State Pollution Control Board (SPCB) Hazardous and Other Wastes Rules (2016) via automated Form 10 manifests and certified plant directory registries.

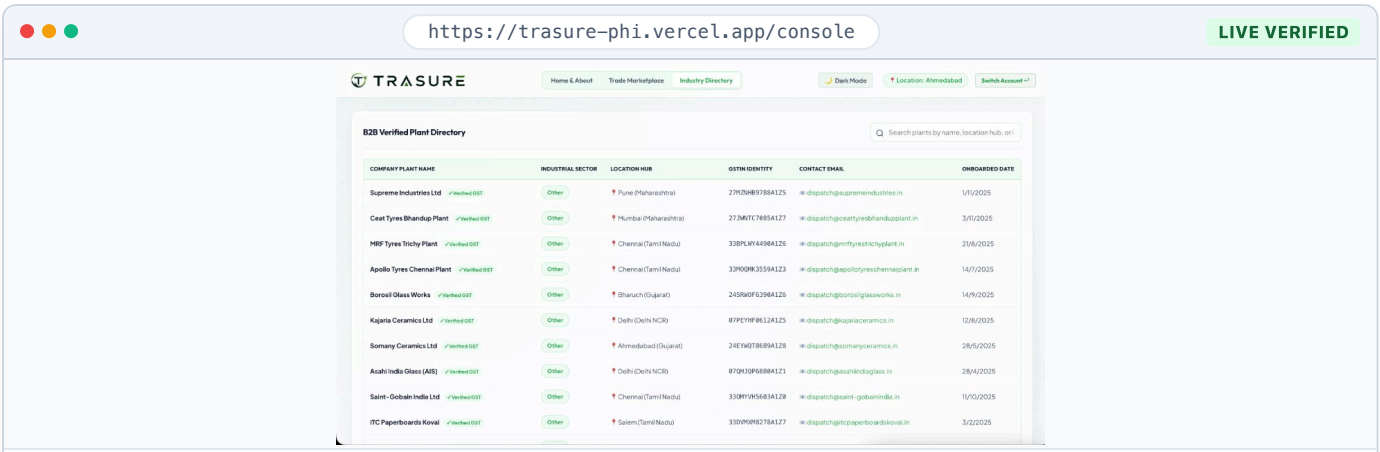


Figure 7: Verified Enterprise Directory — GST and SPCB validated profiles including Supreme Industries, CEAT, MRF, and Borosil.

Form 10 Manifest Protocol

Upon auction award, TRASURE generates a legally binding Form 10 hazardous waste manifest:

```
Manifest Ref: TRS-MFT-2025-0881
SPCB Consent: SPCB/HW/AHM/2024/0149
Hash: e3b0c44298fc1c149afbf4c8996fb92427...
```

Administrative Seed Reset

Authorized campus evaluators can restore the system to its initial baseline state at any time:

- Resets 10 live material listings to verified starting inventories.
- Reinitializes 24-hour reverse auction countdown clocks.
- Purges demo session cookies and clears cached logistics routes.

07 MODULE 7: OPERATIONAL TROUBLESHOOTING & RUNBOOK

Troubleshooting Guide, FAQ & Support Matrix

Operational troubleshooting runbook for plant dispatch managers, logistics leads, and system evaluators encountering edge conditions.

ERROR / SYMPTOM	ROOT CAUSE	RESOLUTION PROTOCOL	SEVERITY & ACTION
PIN Authentication Failure	Incorrect 4-digit PIN entered for selected corporate profile.	Verify credentials against master seed list. Default test PIN for all registered profiles is `1234`.	LOW Re-enter PIN
GIS Route Polyline Unavailable	OSRM public routing API timeout or client network restriction.	System automatically engages mathematical Haversine great-circle fallback. Verify internet connection.	LOW Auto Fallback
Auction Outbid Lock	Counter-bid amount is equal to or higher than current leading bid.	Enter a bid strictly lower than the displayed *Current Leading Bid*. Use quick-bid buttons `–₹50` or `–₹100`.	OPERATIONAL Adjust Bid
Timer Anti-Sniping Loop	Repeated bids within 60s window continuing to trigger overtime.	Normal system behavior designed to ensure competitive price discovery. Wait for bidding cessation.	DESIGN FEATURE Intended Flow
SPCB Validation Rejection	Plant profile SPCB hazardous consent registration expired.	Upload renewed consent letter in Plant Directory settings or re-select SPCB-compliant profile.	COMPLIANCE Renew SPCB

Official Academic Escalation Desk

For academic verification, evaluation queries, or live platform technical support, contact the project author directly:

Lead Engineer: Niv J. Sorathiya **Roll Number:** 24BCB037 (CSBS) **Institution:** PDEU Gandhinagar

College Email: 24bcb037@sot.pdpu.ac.in **Personal Email:** niv.sorathiya@gmail.com **Live URL:** trasure-phi.vercel.app

Evaluation Summary: Project TRASURE represents a fully functional, production-deployed Software Engineering deliverable meeting all curricular criteria for Course 20CS301T / 20CS301P at Pandit Deendayal Energy University.